

COMPARISON OF EFFICACY BETWEEN EDARAVONE AND RILUZOLE COMBINATION THERAPY FOR PATIENT WITH AMYOTROPHIC LATERAL SCLEROSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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ABSTRACT

Background: Amyotrophic Lateral Sclerosis (ALS) is a fatal neuromuscular disease with a survival period of less than 5 years. Although the past decade has shown a major growth of interest in Edaravone research due to its superior efficacy, a growing number of research done on Riluzole combinations for ALS therapy.

Objective: A systematic review is needed to compare the patient outcomes as shown in the ALS functional rating scale (ALSFRS-R) of Edaravone and Riluzole combinations.

Methods: This research is a systematic review from PubMed, ProQuest, and Science Direct. The studies included were randomized controlled trials (RCT) and post-hoc analysis of RCT published from 2012 to 2023. This research included 16 studies (11 RCT studies and 5 post-hoc studies of RCT). The studies discussed the patient population, side effects, and ALSFRS-R scores.

Results: Edaravone showed better efficacy than Riluzole combination in maintaining ALSFRS-R score with mean difference between Edaravone (MD=2.19, 95% CI: 0.42-3.96, p=0.02) and Riluzole (MD=1.67, 95% CI: -0.19-3.52, p=0.08). The meta-analysis showed heterogeneity in both Edaravone (I²=90%, p<0.00001) and Riluzole (I²=77%, p=0.004) studies.

Conclusion: The results conclusively showed that Edaravone was more efficacious than the explored Riluzole combinations so far. Nevertheless, Riluzole-Masitinib showed promising results to be further explored.

Keywords: Edaravone, Riluzole, Amyotrophic Lateral Sclerosis, ALSFRS-R

Introduction

Amyotrophic Lateral Sclerosis (ALS) is a neurodegenerative disease that generally affects the body's propulsion systems, but this disease can also cause symptoms outside of these system disorders.¹ Worldwide prevalence of this disease reaches 4.42 per 100,000 population and the incidence of this disease is reaching 1.59 per 100,000 patients per year.² The death rate of ALS sufferers is estimated to reach 30,000 deaths each year with an average time of survival of only two to nine years after the disease is diagnosed.^{2,3}

Currently the therapy that has been approved by the United States and Europe for the treatment of ALS is only one drug, namely Riluzole. However, this drug has only shown minimal efficacy in the treatment of patients with ALS in several studies. This drug can extend the time that ALS sufferers survive without a tracheostomy for two to three months.^{4,5} Since Riluzole was first approved for the treatment of ALS patients in 1995, there have been several studies conducted to create new drugs that can be used to treat ALS patients. In the last twenty years there have been

sixty molecules that have been tested as potential therapies for ALS sufferers, but from these various studies there have been no new drugs that have provided satisfactory results for ALS therapy.^{3,6,7}

From various studies that have been conducted, one of the drug molecules that is quite promising in ALS therapy is Edaravone.^{3,8} Therefore this systematic review was made to assess the efficacy of therapy using Edaravone in ALS patients when compared to various drugs that have been carried out in clinical trials between 2012 and 2023.

Methods

Study Design

This study was made based on the guidelines for writing a systematic review from "Preferred Reporting Items for Systematic Reviews and Meta-Analysis" (PRISMA). ALS sufferers.⁹ This systematic review was designed to determine the difference in the level of effectiveness between the use of Edaravone compared to the combination of Riluzole in maintaining the ALSFRS-R score in patients with ALS.

Eligibility Criteria

Studies included in this study had to meet several requirements, namely being an experimental randomized controlled trial or a post-hoc study based on the results of a randomized controlled trial, written in English, the research subjects being patients with ALS and using Edaravone or a combination of Riluzole as therapy. Research studies would be excluded if the journal obtained was not a complete available article or was still the protocol of a randomized controlled trial study. In addition, duplicate publications, observational studies, review articles, and case reports would also be excluded from this study.

Search Strategy

The studies included in this systematic review were obtained through three databases from Pubmed, ScienceDirect, and ProQuest. The study articles taken were journals published between 2012 and 2023 and were written in English. The keywords used in the search for sources for this systematic review were “Amyotrophic Lateral Sclerosis”, “Edaravone”, “Riluzole”, and “Randomized Controlled Trial”. These keywords were used as a combination of keywords in the database used to search for relevant literature.

Study Selection

Journal article screening was carried out independently by three authors. If there were differences of opinion regarding the included journals, a settlement would be carried out through discussion between the authors. The inclusion criteria used were the population of patients with ALS, therapy in the form of Edaravone or a comparator, namely a combination of Riluzole, and clinical outcomes in the form of the ALSFRS-R score.

Data Extraction and Bias Risk Assessment

The data extracted from these journals were the characteristics of the patient population, side effects, and clinical outcomes of therapy, which mainly consist of changes in the ALSFRS-R score. The quality assessment of the included journals was carried out by three reviewers. If there was disagreement in the assessment of the journal, it will be discussed between the three reviewers until an agreement is reached. The assessment of bias in this study was based on a scoring system developed by the Cochrane Collaboration. The bias assessment carried out included randomization of the order of participants, allocation secrecy, blinding, incomplete data, selective reporting, and other biases. The quality of the study would be categorized into three levels, namely low, unclear, and high.

Statistical Analysis

The data obtained would be meta-analyzed and analyzed using the RevMan 5.4.1 application. Assessment of study heterogeneity was carried out by looking at the I^2 test scores with the interpretation of the results considered homogeneous if the I^2 value $\leq 40\%$. Making forest plots in this study used the Inverse Variance statistical method for continuous data. The model analysis carried out in this meta-analysis used a random effect model if the results of the forest plot touch or cross the zero threshold of the effect line. For the effect size model of this study, we used differences in means for assessment of the ALSFRS-R

outcome scores and differences in risk for assessment of therapy with 95% confidence intervals (CIs).

Results

In this systematic review, sixteen studies were included (Figure 1). The study consisted of eleven randomized controlled trials and five post-hoc studies that were conducted using data from randomized controlled trials published between 2012 and 2022 for a total population of 1982 people. Demographic characteristics of the study (Table 1) and side effects of therapy (Table 2) could be seen in the table.

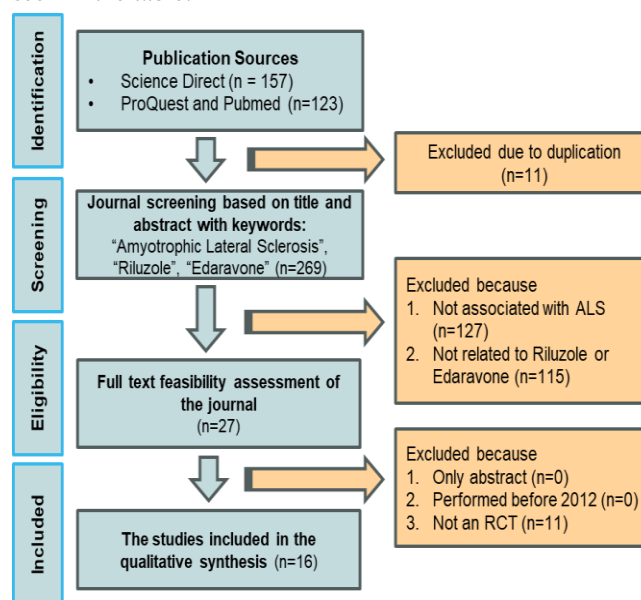
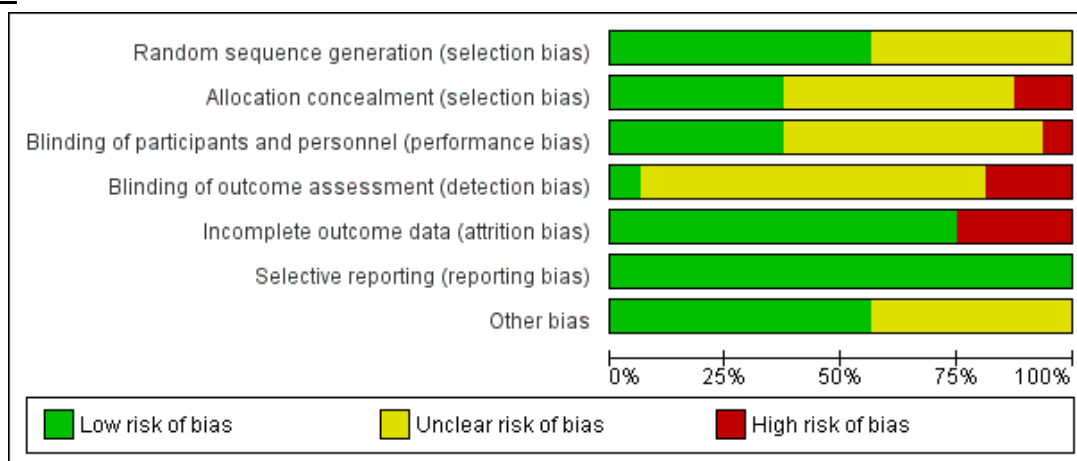


Figure 1. PRISMA Flowchart

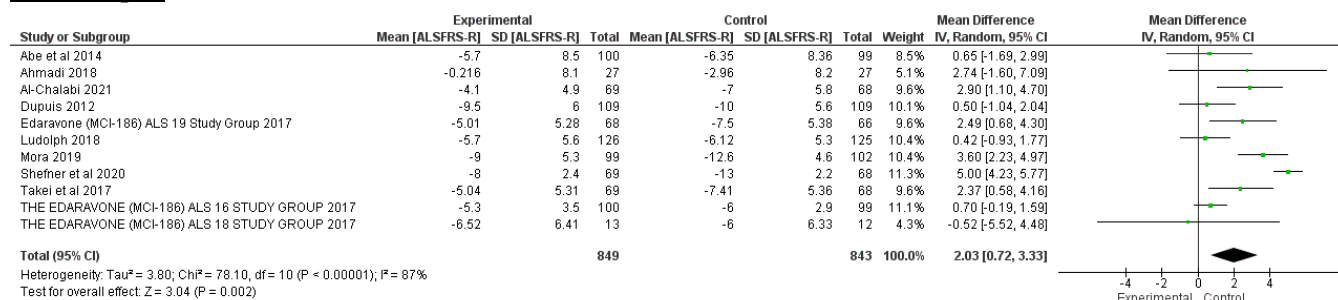
Based on the assessment of the risk of bias (Figure 2), the assessment of the first four indicators showed that most studies have a low risk of bias, except for a few post-hoc studies that cannot be reviewed for these four indicators.11–26 Exceptions were found in three studies, namely research by Mora et al. and the writing group on behalf of the Edaravone (MCI-186) ALS 17 and 18 study group, where there was no disguised allocation in the first study, no disguised allocation and disguised participants and personnel in the second study, and no incognito assessment of the results in the third study.14,15,19 On the fifth indicator, namely incomplete data results, there were four studies, namely the writing group on behalf of the Edaravone (MCI-186) ALS 19 study group, Mora et al., Abe et al., and Kovalchuk et al. who reported missing data.13,16,18,19 Thus the four studies were considered to have a high risk for incomplete data. The last two indicators, namely the reporting of selective results and other biases, all studies were classified as low risk, except for a few post-hoc studies which could not be assessed using these indicators.22–26

Of the sixteen studies included in this study, there were eleven journals that could be meta-analyzed. Of the eleven studies that underwent meta-analysis, all studies used the ALSFRS-R score to assess the efficacy of therapy.11–21 The ALSFRS-R scores for all studies were reported using the least-squares mean difference method.

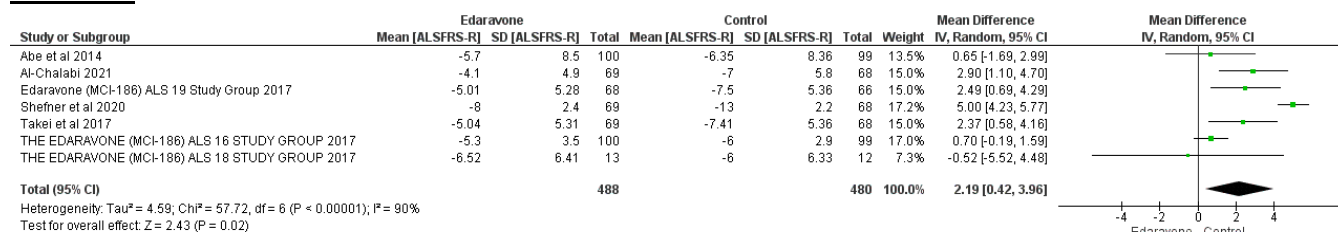
Risk of Bias



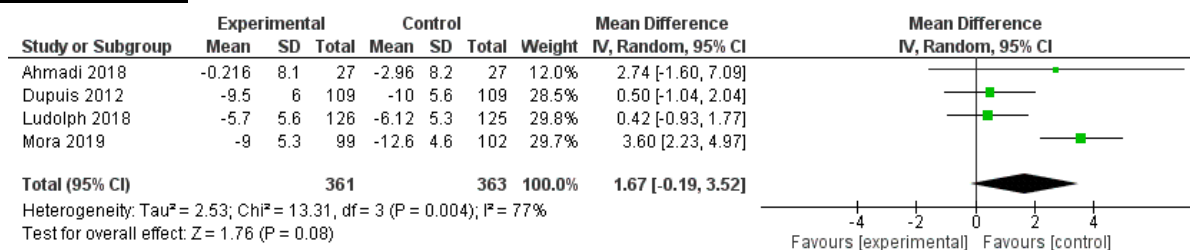
All Therapies



Edaravone



Riluzole Combination



Funnel Plot

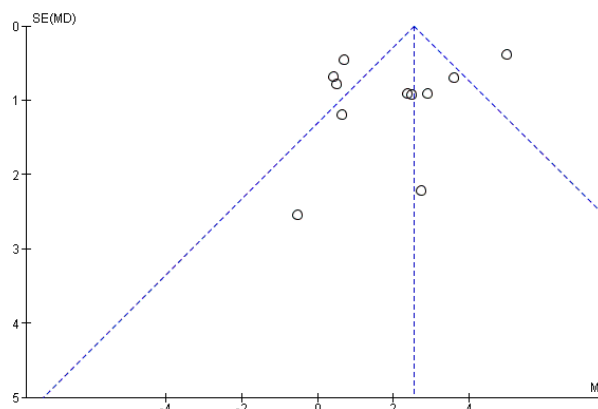


Figure 2. Bias Risk Analysis and Meta-analysis

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Authors		Ahmadi et al		Mora et al		Koji et al		Koji et al		Takahashi et al		The Edaravone (MCI-186) ALS 16 Study Group				Shefner et al	
Year		2018		2019		2017		2017		2017		2017				2019	
Country		Iran		Spain, Argentina, Italy, Slovakia, Netherlands, Canada, Portugal, Mexico, Greece		Japan		Japan		Japan		Japan				Japan	
Study Design		pilot randomized controlled trial		randomized controlled trial		post-hoc study of randomized controlled trial		post-hoc study of randomized controlled trial		post-hoc study of randomized controlled trial		post-hoc study of randomized controlled trial				post-hoc study of randomized controlled trial	
Jadad score		5		3		3.5		3		3.5		3				3	
Drugs		Nanocurcumin		Placebo		Riluzole-Placebo		Riluzole-Masitinib 4.5		Riluzole-Masitinib 3.0		Edaravone		Placebo		Edaravone	
Population		(n=27)		(n=27)		(n=133)		(n=130)		(n=131)		(n=69)		(n=68)		(n=65)	
Sex																	
Male		21 (78%)		18 (67%)		80 (60.2%)		83 (63.8%)		81 (61.8%)		12 (54.5)		13 (81.3)		18 (62.1)	
Female		6 (22%)		9 (33%)		53 (39.8%)		47 (36.2%)		50 (38.2%)		10 (45.5)		3 (18.8)		11 (37.9)	
Age, mean		51.5 ± 13.1		58.5 ± 9.6		55.2 ± 10.6		55.5 ± 10.6		55.7 ± 10.2							
<65 years old		-		-		-		-		-							
≥65 years old		-		-		-		-		-							
Age at onset appearance		49.2 ± 13.1		56.2 ± 9.5		-		-		-							
Body weight, Kg		-		-		-		-		-		55.0 (40-72)		64.0 (50-77)		57.0 (37-83)	
Height, cm		-		-		-		-		-							
BMI, kg/m ²		25.4 ± 3.9		23.7 ± 4.5		-		-		-							
Diagnosis ALS																	
sporadic		-		-		-		-		-		21 (95.5)		16 (100.0)		29 (100.0)	
familial		-		-		-		-		-		1 (4.5)		0		0	
ALS Diagnostic Criteria (El Escorial)																	
definite		-		79 (59.4%)		76 (58.5%)		73 (55.7%)				10 (45.5)		7 (43.8)		9 (31.0)	
probable		-		43 (32.3%)		44 (33.8%)		43 (32.8%)				12 (54.5)		9 (56.3)		20 (69.0)	
laboratory support possible		-		11 (8.3%)		10 (7.7%)		15 (11.5%)									
suspected		-		-		-		-									
ALS Severity (Japan ALS Severity Classification)																	
Grade 1		-		-		-		-									
Grade 2		-		-		-		-									
Grade 3		-		-		-		-									
Disease duration, years		-		-		-		-									
Disease duration, months		24 (8-60)		24 (5-84)		18.1 ± 8.6		19.2 ± 9.6		19.2 ± 8.2							
Early symptoms																	
Bulbar		6 (22%)		2 (7%)		24 (18.0%)		23 (17.7%)		21 (16.0%)		4 (18.2)		0 (0.0)		7 (24.1)	
Extremities		21 (78%)		25 (93%)		-		-		-		18 (81.8)		16 (100.0)		22 (75.9)	
Spinal		-		109 (82.0%)		107 (82.3%)		110 (84.0%)									
ALSFRS-R score																	
Before observation		36 ± 8.1		32.6 ± 8.2		38.1 ± 5.5		37.5 ± 5.5		37.4 ± 5.7		44.5 (37-48)		44.5 (39-47)		44.0 (41-48)	
After 12-weeks observation		-		-		-		-		-		42.5 (36-47)		43.5 (37-46)		42.0 (37-47)	
Changes after the observation period																	
-4 or -3		-		-		-		-		-							
-2 or -1		-		-		-		-		-							

Table 2. Drug Side Effects Based on Each Studies

Authors, Year	The Writing Group on behalf of the Edaravone (MCI-186) ALS 19 Study Group				Ludolph et al		The Writing Group on Behalf of the Edaravone (MCI-186) ALS 19 Study Gorup				Ahmadi et al		Mora et al			The Writing Group on Behalf of the Edaravone (MCI-186) ALS 18 Study Group				The Writing Group on Behalf of the Edaravone (MCI-186) ALS 17 Study Group				Abe et al		Kalin et al		Dupuis et al	
	2017		2018		2017		2018		2019			2017		2017				2014		2017		2012							
Drugs	Edaravone	placebo	Rasagiline	Placebo	Edaravone-Edaravone	Placebo-Edaravone	nanocurcumin	placebo	Riluzole-Placebo	Riluzole-Masitinib 4.5	Riluzole-Masitinib 3.0	Edaravone	Placebo	Edaravone-Edaravone	Edaravone-Placebo	Placebo-Edaravone	Placebo	Edaravone	Edaravone	Placebo	Riluzole-Pioglitazone	Riluzole-placebo							
	n=	n=			n=65	n=58			n=133	n=129	n=131	n=12	n=12	n=48	n=44	n=88	n=104	n=101	n=184	n=184	n=109	n=109							
Iron-deficiency anemia									2 (1.5%)	9 (7%)	3 (2.3%)								0 (0%)	4 (2.2%)									
Anxiety									1 (0.8%)	8 (6.2%)	5 (3.8%)																		
Athralgia					4 (6.2%)	1 (1.7%)																							
Cough									3 (2.3%)	7 (5.4%)	0 (0.0%)			0 (0%)	3 (6.7%)	2 (2.3%)					7 (6.4%)	5 (4.6%)							
Bronchitis														0 (0%)	0 (0%)	5 (5.7%)					14 (12.8%)	13 (11.9%)							
Depression			7 (6%)	12 (10%)										1 (2.1%)	3 (6.7%)	3 (3.4%)													
Contact dermatitis	8 (12%)	3 (4%)																	11 (6%)	6 (3.3%)									
Diarrhea	2 (3%)	4 (6%)							5 (3.8%)	10 (7.8%)	11 (8.4%)	3 (23.1%)	1 (8.3%)	3 (6.3%)	1 (2.2%)	6 (6.8%)			8 (4.3%)	9 (4.9%)									
Dysphagia	8 (12%)	10 (15%)	32 (25%)	24 (19%)	6 (9.2%)	13 (22.4%)						2 (15.4%)	0 (0%)	9 (18.8%)	7 (15.6%)	22 (25.0%)	12 (11.5%)	8 (7.8%)	18 (9.8%)	21 (11.4%)									
Diskizia												0 (0%)	3 (25%)																
Dislalia																													
Dyspepsia									4 (3.0%)	9 (7.0%)	3 (2.3%)					6 (6.8%)					32 (29.4%)	34 (31.2%)							
Dyspnea			21 (17%)	17 (14%)					3 (2.3%)	10 (7.8%)	3 (2.3%)	1 (7.7%)	0 (0%)	3 (6.3%)	3 (6.7%)	4 (4.5%)					35 (32.1%)	31 (28.4%)							
Palpebra edema									0 (0.0%)	4 (3.1%)	0 (0.0%)																		
Peripheral edema									1 (0.8%)	9 (7%)	7 (5.3%)										26 (23.9%)	18 (16.5%)							
Excoriation																													
Eczema	5 (7%)	2 (3%)			5 (7.7%)	2 (3.4%)								3 (6.3%)	3 (6.7%)	3 (3.4%)	2 (1.9%)	7 (6.9%)	5 (2.7%)	3 (1.6%)									
Eritema														2 (4.2%)	3 (6.7%)	2 (2.3%)			5 (2.7%)	3 (1.6%)									
Pharyngitis																			5 (2.7%)	5 (2.7%)									
Abnormal liver function																			2 (1.1%)	5 (2.7%)									
Respiratory failure			25 (20%)	31 (25%)	1 (1.5%)	3 (5.2%)			5 (3.8%)	13 (10.1%)	9 (6.9%)	1 (7.7%)	0 (0%)	6 (12.5%)	2 (4.4%)	6 (6.8%)	16 (15.4%)	20 (19.6%)	2 (1.1%)	5 (2.7%)	33 (30.3%)	31 (28.4%)							
Gait disturbance												4 (30.8%)	1 (8.3%)	14 (29.2%)	9 (20.0%)	32 (36.4%)			23 (12.5%)	17 (9.2%)									
Speech disturbance	1 (1%)	2 (3%)										0 (0%)	2 (16.7%)						5 (2.7%)	4 (2.2%)									
Joint ligament disorders												1 (7.7%)	2 (16.7%)	6 (12.5%)	5 (11.1%)	12 (13.6%)			4 (2.2%)	6 (3.3%)									
Musculoskeletal disorders																			4 (2.2%)	6 (3.3%)									
Respiratory disorders	3 (4%)	2 (3%)			4 (6.2%)	4 (6.9%)								3 (6.3%)	1 (2.2%)	3 (3.4%)	3 (2.9%)	6 (5.9%)	8 (4.3%)	2 (1.1%)									
Glucosuria																			7 (3.8%)	3 (1.6%)									
Sleep disorders																					6 (5.5%)	8 (7.3%)							
Saliva hypersecretion																					6 (5.5%)	8 (7.3%)							
Hypoventilation			7 (6%)	4 (3%)																	9 (8.3%)	3 (2.8%)							
Urinary tract infection									2 (1.5%)	6 (4.7%)	2 (1.5%)																		
Upper airway inflammation	5 (7%)	2 (3%)			4 (6.2%)	1 (1.7%)			1 (0.8%)	5 (3.9%)	2 (1.5%)	3 (23.1%)	1 (8.3%)	3 (6.3%)	1 (2.2%)	4 (4.5%)			6 (3.3%)	3 (1.6%)									
Insomnia	5 (7%)	4 (6%)												5 (10.4%)	3 (6.7%)	7 (8.0%)	10 (9.6%)	9 (8.8%)	14 (7.6%)	15 (8.2%)									
Fall					13 (10%)	20 (16%)			7 (5.3%)	12 (9.3%)	9 (6.9%)										11 (10.1%)	11 (10.1%)							
Muscle weakness														1 (2.1%)	3 (6.7%)	5 (5.7%)	9 (8.7%)	7 (6.9%)	8 (4.3%)	10 (5.4%)									
Muscle spasm																					7 (6.4%)	7 (6.4%)							
Constipation	8 (12%)	8 (12%)	6 (5%)	7 (6%)	7 (10.8%)	8 (13.8%)			2 (1.5%)	6 (4.7%)	2 (1.5%)	2 (15.4%)	0 (0%)	10 (20.8%)	8 (17.8%)	20 (22.7%)	17 (16.3%)	13 (12.7%)	23 (12.5%)	24 (13%)	15 (13.8%)	11 (10.1%)							
Contusio	13 (19%)	9 (13%)	10 (8%)	7 (6%)	6 (9.2%)	6 (10.3%)								3 (6.3%)	7 (15.6%)	8 (9.1%)	5 (4.8%)	12 (11.8%)	27 (14.7%)	16 (8.7%)									
Laseration																			2 (1.1%)	4 (2.2%)									
General weakness																					12 (11.0%)	14 (12.8%)							
Myalgia	4 (6%)	1 (2%)										0 (0%)	2 (16.7%)						4 (2.2%)	2 (1.1%)									
Nausea																			4 (2.2%)	1 (0.5%)	6 (5.5%)	5 (4.6%)							
Nasopharyngitis	3 (4%)	5 (7%)	9 (7%)	10 (8%)	6 (9.2%)	4 (6.9%)						2 (15.4%)	2 (16.7%)	10 (20.8%)	12 (26.7%)	28 (31.8%)	22 (21.2%)	22 (21.6%)	27 (14.7%)	29 (15.8%)	13 (11.9%)	12 (11.0%)							
Pain																					6 (5.5%)	5 (4.6%)							
Musculoskeletal pain																			2 (1.1%)	5 (2.7%)	9 (8.3%)	7 (6.4%)							
Extremities pain																					10 (9.2%)	2 (1.8%)							
Upper abdominal pain									3 (2.3%)	8 (6.2%)	4 (3.1%)																		
Low back pain																													
Back pain	4 (6%)	1 (2%)	3 (2%)	8 (6%)																									
Increased blood bilirubin									1 (0.8%)	6 (4.7%)	0 (0.0%)								7 (3.8%)	7 (3.8%)	9 (8.3%)	10 (9.2%)							
Loss of body weight									9 (6.8%)	14 (10.9%)	13 (9.9%)										18 (16.5%)	9 (8.3%)							
Subcutaneous hemorrhage																													
Pyrexia														3 (6.3%)	1 (2.2%)	2 (2.3%)			3 (1.6%)	4 (2.2%)									
Aspiration pneumonia		0 2 (3%)	9 (7%)	8 (6%)	2 (3.1%)	3 (5.2%)													1 (0.5%)	4 (2.2%)									
Pruritus					5 (7.7%)	1 (1.7%)						2 (15.4%)	1 (8.3%)	1 (2.1%)	1 (2.2%)	8 (9.1%)			0 (0%)	4 (2.2%)	7 (6.4%)	5 (4.6%)							
Dizziness																			0 (0%)	7 (3.8%)									
Rash			7 (6%)	2 (2%)	4 (6.2%)	2 (3.4%)			6 (4.5%)	10 (7.8%)	3 (2.3%)	2 (15.4%)	0 (0%)						3 (1.6%)	4 (2.2%)	8 (7.3%)	4 (3.7%)							
Maculopapular rash									1 (0.8%)	11 (8.5%)	6 (4.6%)								7 (3.8%)	4 (2.2%)									
Generalized rash									0 (0.0%)	4 (3.1%)	1 (0.8%)																		
Stomatitis												0 (0%)	3 (25.0%)						3 (1.1%)	8 (4.3%)									
Headache	4 (6%)	5 (7%)	9 (7%)	8 (6%)	1 (1.5%)	3 (5.2%)						3 (23.1%)	2 (16.7%)				3 (2.9%)	8 (7.8%)	15 (8.2%)	10 (5.4%)	7 (6.4%)	7 (6.4%)							
Bacterial cystitis									2 (1.5%)	6 (4.7%)	3 (2.3%)																		
Tinea pedis														3 (6.3%)	3 (6.7%)	4 (4.5%)			4 (2.2%)	2 (1.1%)									
Pelvic vein thrombosis												0 (0%)	1 (8.3%)																
Vertigo			9 (7%)	6 (5%)																									

Based on the analysis of changes in the ALSFRS-R score from the start of therapy to the end of treatment for all types of treatment (Figure 2), the results were heterogeneous ($I^2=87\%$, $p<0.00001$).

In this analysis, because there were values that crossed the zero value of the effect line, random assessments were carried out, so that significant results were obtained that the experimental model using Edaravone or the Riluzole combination showed a decrease in the ALSFRS-R score which was less than the control ($MD=2.03$, 95 % CI: 0.72-3.33, $p=0.002$). Further subgroup analysis was then performed to assess whether there was a difference in therapy using Edaravone in combination with Riluzole.

From the analysis conducted in the Edaravone therapy group on changes in the ALSFRS-R score (Figure 2), the results were still heterogeneous ($I^2=90\%$, $p<0.00001$). Analysis of the Edaravone group was also carried out randomly. The results of a meta-analysis conducted on the Edaravone group alone showed better results in maintaining the ALSFRS-R score than the control ($MD=2.19$, 95% CI: 0.42-3.96, $p=0.02$).

The analysis performed on the Riluzole combination group on changes in the ALSFRS-R score (Figure 2) still showed heterogeneous results ($I^2=77\%$, $p=0.004$). Forest plot analysis of this subgroup was also carried out using a random analysis model, but the results obtained from the Riluzole combination were not significant ($MD=1.67$, 95% CI: -0.19-3.52, $p=0.08$).

A funnel plot covering all studies using Edaravone (Figure 2) as an intervention indicated the possibility of publication bias in this sample. Most studies lied within the standard error of 0–2. Qualitative analysis of the funnel plots revealed an asymmetry that indicated a bias in the results of the studies. This was most likely due to the small sample sizes in these studies because ALS was such a rare disease that studies with larger populations were difficult to conduct. A similar analysis was not performed for Riluzole due to the very small number of studies (four studies), making interpretation difficult and the possibility of ambiguity high.

Discussion

The results of a meta-analysis of clinical trials using Edaravone showed better efficacy compared to the Riluzole combination.^{11–21} In studies using the Riluzole combination, only the combination with Masitinib showed significant results.¹⁹ Masitinib itself is a therapy working by inhibiting tyrosine kinase so that mast cells and neutrophils are reduced.^{27,28} This prevents degeneration and damage to neurons in the spinal cord.^{27,29}

In ALS itself, one of the pathological mechanisms that plays an important role is oxidative stress.³⁰ Reactive oxygen species (ROS) is one of the enzymes associated with oxidative stress and causes inflammation and damage to cells due to disturbances in lipids and DNA.^{30,31} Edaravone is a treatment that works as one of the scavengers of oxidative stress, so it can prevent neurodegenerative processes in ALS patients.^{32,33}

There have been several studies on the safety of Edaravone itself.^{11,13,15,16,34–37} There was no significant difference

between the side effects experienced by Edaravone recipients and placebo. Most of the side effects that occurred were due to the development of ALS disease and more serious side effects were experienced by patients who received a placebo. However, there is data showing higher side effects of Edaravone on skin tissue with the most common side effects being eczema, contact dermatitis, rash, and erythema, and no harmful side effects were found on the skin.^{11,13,15,16} Further research regarding the safety of Edaravone after three years of obtaining approval from the American Food and Drug Administration (FDA) had also been carried out. The results of this study showed that Edaravone did not provide additional side effects compared to previous studies and control trials.³⁴ Studies on the long-term use of Edaravone orally and intravenously had also demonstrated the safety of this therapy.^{35–37}

Evaluation of discontinuation and change of therapy to Edaravone had been carried out. In this study, there were three treatment groups, namely ALS patients who had received Edaravone and then continued with Edaravone (E-E), placebo recipients then continued with Edaravone (P-E), and Edaravone recipients then continued with placebo (E-P). The results of this study indicated that E-E recipients gave the best results and the results of the E-P group were better than the P-E group in terms of changes in the ALSFRS-R score. In addition, early initiation of Edaravone had been shown to provide better benefits in maintaining the functional status of ALS patients.^{14,24,25}

Several follow-up studies discussed the analysis of sub-populations receiving Edaravone which were grouped into Efficacy-expected sub-population (EESP) and definite/probable EESP 2 years (dpEESP2y). The EESP population was a sub-population with a score of ≥ 2 on all ALSFRS-R instruments with a forced vital capacity (FVC) of more than 80%. Meanwhile, the dpEESP2y population was an EESP population diagnosed with El Escorial criteria and had suffered from ALS for ≤ 2 years. In that study, the EESP and dpEESP2y sub-populations showed better ALSFRS-R scores compared to the overall analysis group, with the best results and the fewest side effects found in the dpEESP2y group.^{14,22,24}

Various studies had shown good results regarding the use of Edaravone in ALS patients.^{11–16,22–26,34–37} However, most of the existing studies had been conducted in Japan.^{11–16,22–26} Research on Edaravone outside Japan itself had started to be carried out even though there were still few.^{35–37} Therefore, further research on the use of Edaravone on other ethnicities, early initiation of Edaravone, and research on certain sub-populations are interesting to be carried out. In addition, further research on the use of Masitinib as a combination therapy for ALS patients also needs to be further explored.

Conclusion

The results conclusively showed that Edaravone was more efficacious than the explored Riluzole combinations so far. Nevertheless, Riluzole-Masitinib showed promising results to be further explored.

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